

CO2- ASSESSMENT TOOL 2

Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System

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Scenario-Based Problem Analysis

Problem Statement

The hospitality industry has undergone rapid digital transformation due to the increasing expectations of customers, global competition, and technological advancements in Artificial Intelligence (AI), Internet of Things (IoT), cloud computing, and mobile applications. Modern hotel guests expect seamless digital services such as online booking, contactless check-in, personalized room preferences, intelligent recommendations, and instant customer support. However, many hotels continue to rely on traditional hotel management systems that focus only on reservations, billing, and basic customer records.

Traditional hotel operations involve extensive manual coordination between reception, housekeeping, maintenance, restaurant, and management departments. These manual processes often result in long waiting times, inefficient room allocation, delayed housekeeping services, poor maintenance management, inconsistent customer experiences, high operational costs, and underutilization of hotel resources.

To overcome these limitations, an Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System is proposed. The system integrates AI, IoT devices, predictive analytics, and cloud-based services into a unified hotel management platform. It automates operational workflows, predicts guest requirements, optimizes resource utilization, supports dynamic room pricing, enables predictive maintenance, monitors energy consumption, and provides personalized guest experiences through intelligent recommendations and chatbot assistance.

Business Domain

Hospitality and Hotel Management Industry

Existing Challenges

- Manual and time-consuming check-in/check-out procedures.

- Inefficient room allocation during peak occupancy.
- Lack of personalized guest recommendations.
- Delays in housekeeping and maintenance requests.
- Poor communication between hotel departments.
- High energy consumption due to inefficient monitoring.
- Difficulty in forecasting room demand and occupancy.
- Inadequate data analytics for managerial decision-making.
- Limited customer engagement after booking.
- Increased operational costs due to resource wastage.

Stakeholders

- Hotel Guests
- Reception and Front Desk Staff
- Hotel Manager
- Housekeeping Staff
- Maintenance Staff
- Restaurant and Room Service Staff
- Hotel Administration
- AI Recommendation Engine
- Payment Gateway Provider
- Third-Party Travel Platforms
- IoT Device Providers

System Objectives

The proposed system aims to:

- Automate hotel operations using AI and smart technologies.
- Provide contactless and intelligent check-in/check-out.

- Offer personalized room, dining, and activity recommendations.
- Optimize housekeeping and maintenance scheduling.
- Enable dynamic room pricing based on demand prediction.
- Reduce operational costs through energy optimization.
- Improve guest satisfaction through real-time support.
- Provide dashboards and analytics for hotel management.
- Integrate IoT devices for smart room control and monitoring.

The proposed Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System will significantly improve operational efficiency, reduce manual intervention, enhance customer satisfaction, and support data-driven decision-making across all hotel departments.

Software Requirement Specification (SRS)

1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) defines the functional and non-functional requirements of the Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System. It serves as a reference document for developers, testers, project managers, clients, and stakeholders involved in the system development lifecycle.

1.2 Scope of the System

The proposed system provides an integrated AI-powered hotel management solution capable of managing reservations, room allocation, contactless check-in/check-out, housekeeping, maintenance, restaurant services, payments, personalized recommendations, energy monitoring, predictive analytics, and administrative management.

The system can be deployed for:

- Hotels

- Resorts
- Business Hotels
- Luxury Hotels
- Boutique Hotels
- Multi-branch Hotel Chains

1.3 Definitions

- AI: Artificial Intelligence
- IoT: Internet of Things
- PMS: Property Management System
- CRM: Customer Relationship Management
- API: Application Programming Interface
- SRS: Software Requirement Specification

1.4 Intended Users

- Guests
- Receptionists
- Housekeeping Staff
- Maintenance Staff
- Restaurant Staff
- Hotel Managers
- System Administrators
- Hotel Owners

2. Problem Description

Hotels operate multiple departments simultaneously, including reservations, reception, housekeeping, maintenance, restaurant services, and management. Lack of integration between these departments causes delays, duplication of work, and poor customer experiences.

For example:

- A room may remain unavailable because housekeeping status is not updated.
- Maintenance requests may be overlooked due to manual tracking.
- Guests may receive generic offers instead of personalized recommendations.
- Managers may lack real-time occupancy and revenue information.

The proposed AI-based system addresses these issues through automation, intelligent decision support, and real-time data synchronization.

3. Scope of the System

The system includes the following modules:

Reservation Module

- Room availability search
- Booking
- Cancellation
- Modification
- Group reservations

Guest Management Module

- Registration
- Identity verification
- Profile management

- Loyalty program

Smart Room Management

- AI room allocation
- Digital room keys
- IoT-based room control

Housekeeping Module

- Task scheduling
- Cleaning status tracking
- Staff assignment

Maintenance Module

- Complaint management
- Predictive maintenance
- Equipment monitoring

Restaurant Module

- Digital menu
- Room service
- Restaurant reservations

Payment Module

- Online payment
- Invoice generation
- Refund management

Analytics Module

- Occupancy reports
- Revenue analysis
- Customer behavior analyt

4. Functional Requirements

4.1 User Authentication

The system shall:

- Allow guest registration.
- Support login using email, phone number, or social accounts.
- Provide multi-factor authentication.
- Allow password reset.
- Support role-based access control.

4.2 Reservation Management

The system shall:

- Display real-time room availability.
- Allow booking from web and mobile applications.
- Allow reservation modification.
- Allow reservation cancellation.
- Generate booking confirmation.
- Synchronize reservations across hotel branches.

4.3 AI-Based Room Recommendation

The AI engine shall:

- Analyze guest preferences.
- Recommend suitable room types.
- Recommend premium upgrades.
- Suggest seasonal offers.
- Recommend additional services.

4.4 Contactless Check-In

The system shall:

- Verify digital identity.
- Validate reservation.
- Generate digital room key.
- Enable QR-code based check-in.
- Notify reception automatically.

4.5 Contactless Check-Out

The system shall:

- Calculate final bill.
- Display charges.
- Process online payment.
- Generate electronic invoice.
- Deactivate room access automatically.

4.6 Smart Room Allocation

The system shall:

- Assign rooms automatically.
- Consider occupancy optimization.
- Consider guest preferences.
- Avoid maintenance-blocked rooms.
- Minimize room switching.

4.7 Housekeeping Management

The system shall:

- Generate cleaning tasks.
- Assign tasks to staff.
- Update room status.

- Notify reception when cleaning is completed.
- Track housekeeping performance.

4.8 Maintenance Management

The system shall:

- Accept maintenance requests.
- Assign technicians.
- Track maintenance status.
- Generate predictive maintenance alerts.
- Maintain equipment history.

4.9 Restaurant and Room Service

The system shall:

- Display digital menu.
- Accept food orders.
- Recommend meals using AI.
- Track preparation status.
- Track delivery status.

4.10 Payment Processing

The system shall:

- Support UPI, cards, net banking, and wallets.
- Process refunds.
- Generate invoices.
- Maintain payment history.
- Integrate with payment gateways.

4.11 AI Guest Experience

The AI engine shall:

- Recommend restaurants.
- Recommend tourist attractions.
- Recommend hotel facilities.
- Analyze guest feedback.
- Provide chatbot assistance.

4.12 Energy Management

The system shall:

- Monitor electricity usage.
- Monitor HVAC systems.
- Monitor lighting systems.
- Automatically switch off unused devices.
- Generate energy optimization reports.

4.13 Management Dashboard

The system shall:

- Display occupancy rate.
- Display revenue.
- Display room availability.
- Display guest satisfaction score.
- Display housekeeping status.
- Display maintenance status.
- Display AI demand forecast.

4.14 Administration

The administrator shall:

- Manage users.
- Manage roles.

- Configure pricing.
- Configure AI settings.
- Generate reports.
- Maintain audit logs.

5. Non-Functional Requirements

Performance

Requirement	Specification
Reservation Response	Less than 2 seconds
Check-In Processing	Less than 1 minute
Concurrent Users	1,000+
AI Recommendation	Less than 3 seconds
Report Generation	Less than 10 seconds

Security

- HTTPS/TLS encryption
- Encrypted payment information
- Role-based authorization
- Multi-factor authentication
- Secure password hashing
- Audit logging
- Compliance with GDPR and local privacy regulations

Reliability

- 99.9% system availability

- Automatic data backup
- Disaster recovery support
- Transaction consistency
- Fault tolerance for AI services

Usability

- Responsive web interface
- Android and iOS support
- Multilingual interface
- Accessibility support
- Intuitive navigation
- Minimal training requirements

Scalability

- Cloud-native architecture
- Horizontal scaling
- Multi-branch support
- API extensibility

Maintainability

- Modular architecture
- Well-documented APIs
- Configurable business rules
- Easy AI model updates

Availability

- 24/7 operation
- Automatic failover
- Load balancing

- Continuous monitoring

Actors and Use Cases

6. Actors

Primary Actors

- Guest
- Receptionist
- Housekeeping Staff
- Maintenance Staff
- Restaurant Staff
- Hotel Manager
- Administrator

Secondary Actors

- AI Recommendation Engine
- Payment Gateway
- IoT Devices
- Third-Party Travel Platforms
- Notification Service

7. Major Use Cases

Guest

- Register/Login
- Search Rooms
- Book Room
- Check-In
- Check-Out

- Order Food
- Request Housekeeping
- Report Maintenance Issue
- Make Payment
- Receive AI Recommendations
- Submit Feedback

Receptionist

- Manage Reservations
- Verify Guest Identity
- Assign Rooms
- Process Check-In
- Process Check-Out

Housekeeping Staff

- View Assigned Tasks
- Update Room Status
- Complete Cleaning Tasks

Maintenance Staff

- View Maintenance Requests
- Perform Maintenance
- Update Maintenance Status

Restaurant Staff

- Receive Orders
- Prepare Orders
- Update Delivery Status

Hotel Manager

- View Dashboard
- Monitor Occupancy
- Manage Pricing
- Review Feedback
- Generate Reports

Administrator

- Manage Users
- Configure System
- Manage AI Models
- Backup Database
- Audit System Activities

Use Case Diagram

@startuml

left to right direction

actor Guest

actor Receptionist

actor Housekeeping

actor Maintenance

actor Restaurant

actor Manager

actor Admin

actor AI

actor PaymentGateway

rectangle SmartHotelSystem {

usecase "Register/Login" as U1

usecase "Book Room" as U2

usecase "Check-In" as U3

usecase "Check-Out" as U4

usecase "Order Food" as U5

usecase "Request Housekeeping" as U6


```
usecase "Report Maintenance" as U7
usecase "Make Payment" as U8
usecase "Get AI Recommendations" as U9
usecase "Manage Reservations" as U10
usecase "Assign Rooms" as U11
usecase "Update Room Status" as U12
usecase "Perform Maintenance" as U13
usecase "Manage Orders" as U14
usecase "View Dashboard" as U15
usecase "Manage Users" as U16
}
```

Guest --> U1

Guest --> U2

Guest --> U3

Guest --> U4

Guest --> U5

Guest --> U6

Guest --> U7

Guest --> U8

Guest --> U9

Receptionist --> U10

Receptionist --> U11

Receptionist --> U3

Receptionist --> U4

Housekeeping --> U12

Maintenance --> U13

Restaurant --> U14

Manager --> U15

Admin --> U16

AI --> U9

PaymentGateway --> U8

@enduml

8. Use Case Descriptions

Use Case 1: Book Room

Actor: Guest

Preconditions: Guest is registered and logged in.

Main Flow:

1. Search available rooms.
2. Select room type and dates.
3. View AI recommendations.
4. Enter guest details.
5. Confirm booking.
6. Make payment.
7. Receive confirmation.

Alternative Flow:

- Payment failure
- Room becomes unavailable
- Invalid dates

Postconditions: Reservation is successfully created.

Use Case 2: Contactless Check-In

Actor: Guest / Receptionist

Preconditions: Valid reservation exists.

Main Flow:

1. Scan QR code.
2. Verify digital identity.
3. Validate reservation.
4. Allocate room.

5. Generate digital key.

Postconditions: Guest receives room access.

Use Case 3: Request Housekeeping

Actor: Guest

Main Flow:

1. Open mobile application.
2. Select housekeeping service.
3. Choose service type.
4. Submit request.
5. Housekeeping staff receives notification.
6. Task is completed.
7. Guest receives confirmation.

Use Case 4: Report Maintenance Issue

Actor: Guest

Main Flow:

1. Select maintenance request.
2. Describe issue.
3. Submit request.
4. Technician is assigned.
5. Repair is performed.
6. Status is updated.

9. Data Requirements

Guest Entity

- GuestID

- Name
- Email
- Phone
- Address
- Nationality
- Preferences
- LoyaltyPoints

Room Entity

- RoomID
- RoomType
- Floor
- Status
- Price
- SmartDeviceStatus

Reservation Entity

- ReservationID
- GuestID
- RoomID
- CheckInDate
- CheckOutDate
- BookingStatus

Payment Entity

- PaymentID
- ReservationID
- Amount

- PaymentMethod
- TransactionStatus

Housekeeping Task Entity

- TaskID
- RoomID
- StaffID
- Status
- CompletionTime

Maintenance Request Entity

- RequestID
- RoomID
- IssueDescription
- Priority
- AssignedTechnician
- Status

Food Order Entity

- OrderID
- GuestID
- RoomID
- MenuItems
- OrderStatus

Feedback Entity

- FeedbackID
- GuestID
- Rating

- Comments
- SentimentScore

10. External Interface Requirements

User Interface

- Guest mobile application
- Guest web portal
- Reception dashboard
- Housekeeping dashboard
- Maintenance dashboard
- Restaurant dashboard
- Manager dashboard
- Administrator dashboard

Hardware Interfaces

- Smart door locks
- Occupancy sensors
- Temperature sensors
- HVAC systems
- Smart lighting
- Energy meters
- QR code scanners

Software Interfaces

- Payment Gateway API
- SMS Gateway
- Email Service
- AI Recommendation Engine

- Chatbot Service
- Third-Party Booking Platforms
- IoT Device Management Platform

Communication Interfaces

- HTTPS
- REST APIs
- WebSockets
- MQTT for IoT communication

11. System Features

- AI-powered room recommendations
- Contactless check-in/check-out
- Digital room keys
- Dynamic room pricing
- Predictive maintenance
- Smart housekeeping scheduling
- Personalized dining recommendations
- Real-time room service
- Energy optimization
- Chatbot guest support
- Loyalty program integration
- Management analytics dashboard
- Multi-branch hotel support
- IoT integration
- Sentiment analysis of guest feedback

12. Assumptions

- Internet connectivity is continuously available.
- Guests possess smartphones.
- IoT devices are installed in hotel rooms.
- Payment gateway services are operational.
- Hotel staff are trained to use the system.
- Historical data is available for AI training.
- Cloud infrastructure is available.

13. Constraints

Technical Constraints

- AI accuracy depends on data quality.
- IoT device compatibility limitations.
- Network dependency.

Operational Constraints

- Existing hotel workflows must be integrated.
- Staff adaptation period is required.
- Continuous monitoring of IoT devices.

Economic Constraints

- Budget limitations.
- Cloud infrastructure costs.
- AI model development expenses.
- IoT installation costs.

Legal Constraints

- Data privacy regulations.
- Secure payment compliance.

- Electronic invoice regulations.
- Data retention policies.

Time Constraints

- Project completion within academic semester.
- Limited development and testing schedule.

14. Requirement Validation

Completeness Check

All stakeholder requirements have been addressed including:

- Reservations
- Payments
- AI recommendations
- Housekeeping
- Maintenance
- Restaurant services
- Energy management
- Analytics
- Administration

Consistency Check

The requirements are logically connected and do not conflict with each other.

Ambiguity Check

Terms such as dynamic pricing, predictive maintenance, and AI recommendation should be further specified during detailed design.

Redundancy Check

Duplicate reservation and room management requirements have been consolidated.

Missing Requirement Analysis

Additional enhancements can include:

- Voice-controlled room automation
- Facial recognition check-in
- Emergency evacuation management
- Carbon footprint monitoring
- Smart parking integration
- Airport transportation booking
- Multilingual AI voice assistant

Conclusion

The Intelligent AI-Based Smart Hotel Operations and Guest Experience Management System provides a comprehensive AI-driven solution for modern hotel management. The system automates operational workflows, enhances guest experiences through personalization, supports predictive decision-making, optimizes resource utilization, and improves communication between hotel departments. By integrating AI, IoT, cloud computing, and analytics, the proposed system significantly increases operational efficiency, reduces manual intervention, lowers operational costs, and improves customer satisfaction. The requirements specified in this SRS provide a complete foundation for the design, development, testing, deployment, and maintenance of the proposed smart hotel management system.